

Findings:

A. Recovery Analysis

The Suez incident triggered a 123-day recovery window, characterized by a 34% surge in "vessel bunching" volume, which evolved into a permanent step-up in global throughput demand. This "new normal" imposes a 1,037-minute efficiency tax per move, exposing a regional disparity where high-volume hubs, such as EMEA, are red-lining due to infrastructure stress, while the AMER hub suffers from structural bottlenecks despite lower throughput. To mitigate the risk of total system seizure during future disruptions, the 2026 investment strategy must prioritize capacity expansion in EMEA to absorb high-intensity surges and a comprehensive process audit in AMER to resolve its inherent operational latency.

B. Allocation Performance

While the global network maintains a 99.84% operational efficiency with a handling time of 1.00 hours per container, the regional and terminal-level data reveal critical infrastructure imbalances that threaten our 2025 goal of 0.85 hrs/container. The Regional Performance analysis identifies a clear disparity: EMEA successfully processes the highest container volumes despite minor inefficiencies, whereas AMER exhibits the most significant process gaps relative to its workload. At the terminal level, our Performance Segmentation highlights a dense cluster of "Bottleneck" and "Underperformer" sites, specifically those in the upper-right quadrant of the scatterplot that are struggling under high workloads, contrasted against "Performers" and "Scalable" hubs that possess the latent capacity to absorb excess volume. With the Priority Status matrix currently flagging nearly all scrutinized terminals as either "Optimal" or requiring a "Process Fix," the immediate operational mandate is to prioritize infrastructure optimization and resource reallocation from high-pressure AMER bottlenecks and EMEA toward our scalable hubs to mitigate the identified 0.16% productivity loss and secure long-term network resilience. To see which terminals can benefit more through capacity expansion, you can access the file to explore my report.

C. Regional Fleet

An analysis of our fleet performance reveals that vessel age and weekend operational lag are the most significant predictors of movement delays, with legacy vessels—which constitute 80% of our fleet (801 units)—performing 1.68 hours slower than modern alternatives. While night shifts surprisingly demonstrate a 1.8% efficiency advantage over day shifts across all regional hubs, this gain is completely offset by a severe 1.5-hour degradation in performance during weekends. The Decomposition Tree highlights the AMER regional hub as the primary operational outlier, driven specifically by a high concentration of Legacy Cargo vessels during day shifts, which peak at an average duration of 521.68 hours. Consequently, our baseline efficiency is structurally tied to a "modernization gap" where the overwhelming throughput concentration of older vessels, particularly in the Passenger and Cargo categories, creates a systemic bottleneck that night-shift gains alone cannot resolve, necessitating a strategic focus on modernizing the AMER-based fleet and addressing the critical weekend service-level drop. To explore more, please access and explore my report.

Recommendations:

- Prioritize EMEA for crane expansions and yard automation. Since this hub bears the brunt of global surges (as seen in the Suez audit), increasing its "peak capacity" is the fastest way to hit the -15% efficiency target.
- Conduct a business process audit specifically for AMER. Since volume isn't the primary cause of delay here (unlike EMEA), the bottleneck is likely administrative, customs-related, or labor-shift synchronization.
- Incentivize Modernization: Offer "Fast-Track" berthing priorities for modern vessels to encourage carriers to upgrade.
- The "Weekend Shift Sync": Apply the high-efficiency night-shift protocols to weekend day-shifts to eliminate the 1.5-hour "service-level sag."
- Overall, to optimize the global network and mitigate the 1,037-minute efficiency tax, the strategic focus must shift toward load balancing across terminals identified as "Performers" or "Scalable." While the EMEA and AMER hubs are currently red-lining due to infrastructure stress and structural bottlenecks, the Allocation Performance scatterplot reveals specific terminals that maintain high efficiency despite varying workloads. By cross-referencing the Regional Fleet analysis with the Priority Status matrix, the team can identify underutilized hubs—particularly in the APAC and LATAM regions—that demonstrate the operational elasticity to absorb excess cargo. Prioritizing these "Scalable" terminals for diverted volumes will alleviate the pressure on high-risk bottlenecks, turning a regional disparity into a strategic advantage and securing the path toward our -15% handling time goal.